

Numerical integration of the Barotropic Vorticity Equation

Federico Giacobbe, Filippo Di Ludovico, Tommaso Camponogara

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1 Introduction

This report aims to reproduce the first successful Numerical Weather Prediction (NWP) performed by Charney, Fjörtoft and von Neumann in 1950. We start from the same set of barotropic vorticity equations they used for the numerical integration.

This set can be derived from the quasi geostrophic system of equations, obtained via a scale analysis of the governing equations for large scale and low frequency variability of the atmospheric state.

One of the main advantages of this approach is that it allows to describe and evolve the entire tridimensional scalar field over time, knowing only its values on a special atmospheric levels, called the "equivalent level", and the physical laws that relates the vertical profile with the known surface. For this reason, the model is referred to as "equivalent barotropic model", or EBM.

Another big step forward was due to the innovative way to numerically solve the Poisson equation on a discretized grid, by carrying out the computations in the frequency domain. These two strategies turned out to play a big role in the success of Charney's effort, together with the availability of the first automatic calculators.

2 Numerical integration

In this section we present the system of equations and we discuss the numerical integration choices that we used for this work.

The set of barotropic QG vorticity equations is shown on the right.

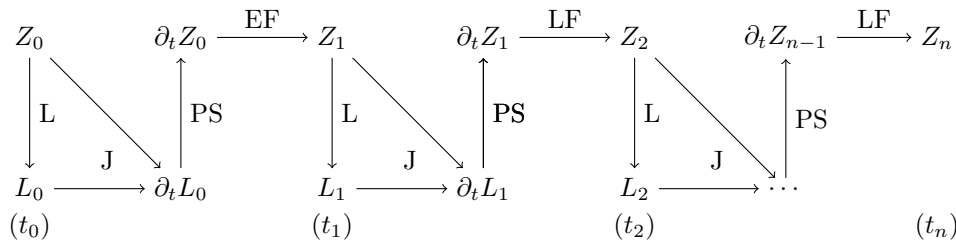
$$\begin{cases} \nabla^2 (\partial_t Z) &= \partial_t L \\ \partial_t L &= J(hL + f, Z) \end{cases}$$

Where $L = \nabla^2 Z = \frac{f}{g} \zeta_g = \frac{f}{g} (\partial_x v_g - \partial_y u_g)$ is a scaled version of the QG vorticity, $J(a, b) = \partial_x a \partial_y b - \partial_y a \partial_x b$ the Jacobian operator, $h = gm^2/f$ a geographic parameter, $f = 2\Omega \sin \phi$ the Coriolis parameter and $m = 2/(1 + \sin \phi)$. Since h is time-independent, both h and m can be evaluated once separately and stored in a fixed "geographic" array, with the same size of the discretization grid, that will be used during the integration to inform the system about the value of the geographic coefficients.

The initial condition Z_0 and the final state Z_{24} are geopotential height fields at 500hPa taken from a reanalysis database for the time window 05/01/1949 3:00 → 06/01/1949 3:00 and they are given as arrays of size 16×19 , with a grid step of approximately 736km. The region represented by the model covers all of North America, Greenland and the North Pole region in general.

Given the initial condition Z_0 , the scaled vorticity at the same time step L_0 is computed as the Laplacian $L_0 = \nabla^2 Z_0$. Since the numerical solution of the latter doesn't provide values of domain boundaries, we extrapolated them with a linear interpolation at first time step and we kept L and Z constant on the borders throughout the integration.

At each time step we computed the time derivative $\partial_t L$ through the Jacobian operator and we obtained $\partial_t Z$ solving the Poisson equation. Then Z is integrated forward in time with a Leapfrog scheme (Euler forward for the initial step) and L at the new time step is computed with the Laplacian $L = \nabla^2 Z$ and the addition of the initial domain boundaries of L_0 . The integration scheme is graphically represented below.



Using this scheme, we integrated the vorticity equations in time using a time step $Dt = 30$ minutes, considering a time window of 24 hours, and we obtained the model result $Z(t_0 + 24h)$.

We compared the forecast field $Z(t_0 + 24h)$ with the final reanalysis field Z_{24} , as shown in Figure 1.

From the graphs some discrepancies between $Z(t_0 + 24h)$ and Z_{24} can be seen. Our model could not provide a good prediction for the final state, represented by Z_{24} , as most of the main features are a bit off their actual place.

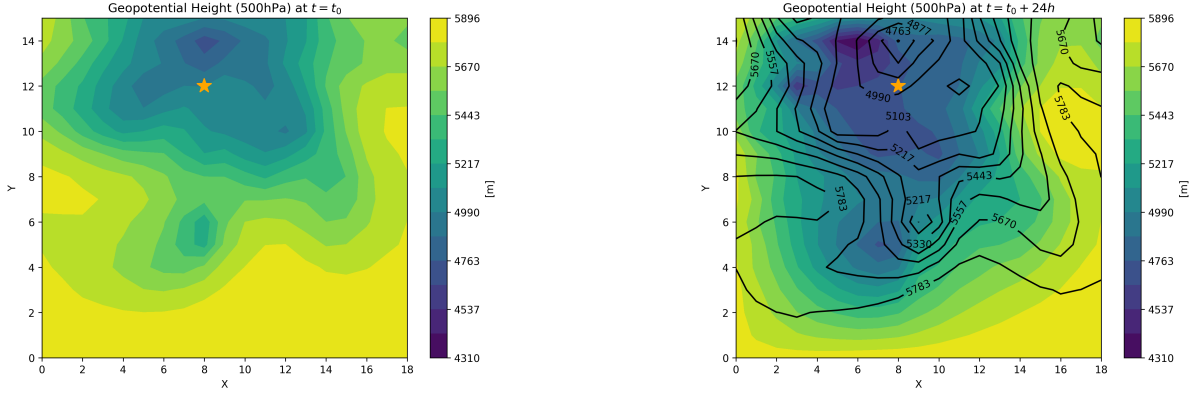


Figure 1: On the left, Z_0 . On the right, $Z(t_0 + 24h)$ contours with superimposed Z_{24} black isolines. The star indicates the North Pole.

Comparing the initial state and the final one, we can also conclude that the simulation deepened and widened the minimum values of the fields. As consequence, the general shape of the field is lost for $Z(t_0 + 24h)$, while for Z_{24} , the general shape is conserved and the evolution of the main features can be better appreciated.

In addition to that, a peculiar behaviour in the prediction of the evolution of the minimums can be noted. The left geopotential trough seems to deepen and to remain in the same position than the initial condition, while in Z_{24} it moves slightly eastward. The polar vortex is evolved with a new geopotential minimum on the left and a general eastward motion around the North Pole, while in Z_{24} the new minimum emerges on the right and the position of the main one seems unchanged.

3 Forecast quality and model skill

To evaluate the forecast quality, we analyzed the field tendency, defined as the 24-hour variation $Z(t_0 + 24h) - Z_0$. Figure 2 compares the modeled tendency against the expected tendency derived from reanalysis data $Z_{24} - Z_0$. The comparison between the two graphs confirms the bad spatial field evolution made by the model, as already stated in section 2. An example is the evolution of the trough, represented by the variations in the central region of the graph. For the model tendency, the "static" deepening is represented by a circular negative variation; for the actual tendency (Fig. 2 right) instead, a pronounced dipolar variation feature suggests the eastward translation of the same trough. The same can be said for the polar vortex minimums, with inverted roles between the model and the observation. Another peculiar and more general discrepancy is that the actual variations in the geopotential field during the 24 hours are generally less pronounced and more localized with respect to the variations predicted by our model. These discrepancy might be partially caused by the coarse horizontal resolution, since the spatial grid spacing is approximately 736km.

Eventually, in the model tendency profile we can also notice that the error along boundaries remains zero throughout the integration and this is due to the fact that we didn't update the geopotential field along them.

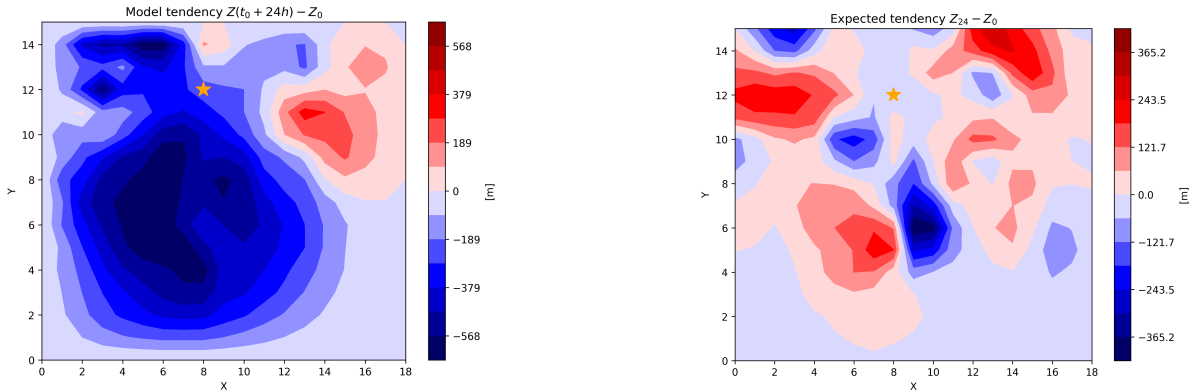


Figure 2: On the left, model tendency, $Z(t_0 + 24h) - Z_0$. On the right, the expected tendency $Z_{24} - Z_0$.

The difference between the model tendency and the expected tendency $Z(t_0 + 24h) - Z_{24}$ (Fig. 3) can be interpreted as the quality of the forecast, painting the value of the cumulated error of our model with respect of Z_{24} , allowing for a more precise and localized analysis of the result.

Its good resemblance of the model tendency confirms what said before about the different magnitude of the variation features between the two tendencies, with a non-local underestimation of the geopotential field by the model. It also suggests that most of the model predicted variations to Z_0 have very little consistency with the observation Z_{24} . We can conclude that the quality of our forecast is bad, as highlighted by this plot mean error, with a value of -194.4 gpm.

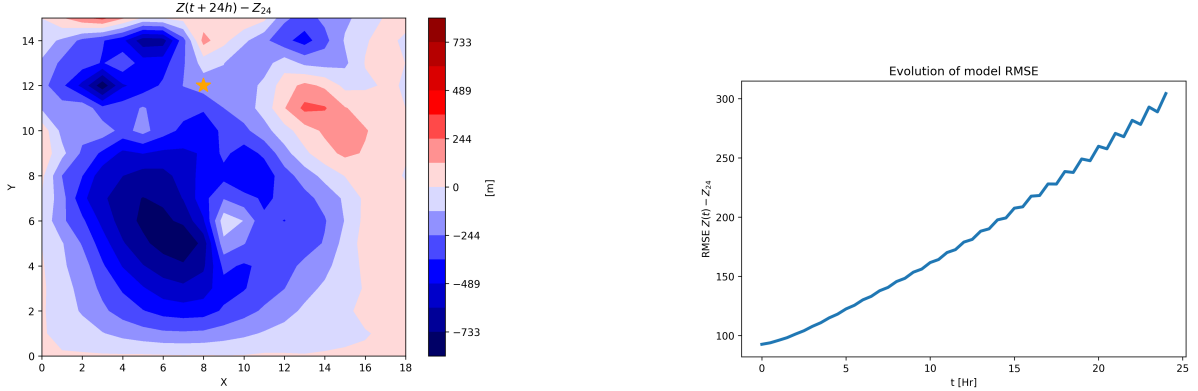


Figure 3: *On the right the comparison between the tendencies $Z(t_0 + 24h) - Z_{24}$, i.e. the forecast quality. On the left the evolution of the model RMSE throughout the time window.*

To quantify the model skill we computed the RMSE of the forecast and compared it against a persistence forecast. We estimated $\text{RMSE}_m = 304.3\text{gpm}$ for the model and $\text{RMSE}_p = 92.6\text{gpm}$ for the persistence. Since RMSE_m exceeds RMSE_p by a factor of three, it follows that the integration causes substantial errors within the 24 hours period. We also noticed an oscillating behaviour in the profile that is superimposed over the main increasing trend. We could not identify its cause, but this oscillation is likely due to the computational scheme.

4 Comments and possible improvements

In this section we try to improve the quality of our forecast model. Starting from the scheme described in section 2, called "Default" in Table 1, we tried to vary independently parameters and methods of integration; the results are shown in Table 1.

First, we decreased the time step by a factor of ten ($Dt = 0.05\text{h}$), then we computed $Z(t_i)$ with Adams–Bashforth 4, a linear multistep method of order 4, that requires four previous step to be applied. They both produce a negligible improvement in the RMSE.

Another way to integrate the system could be to evolve L_i in time with a leapfrog scheme, instead of calculating L_i as the laplacian of Z_i ; due to the little change in RMSE, we concluded that the two methods are equivalent.

Finally, we linearly interpolated both L and Z boundaries at every timestep. The increased RMSE is unexpected, but justifiable due to the better performance of the persistency model.

Due to an oversight, we also found that a better RMSE could be obtained by leaving the boundary of L equal to zero. We have no idea why this happened and cannot give any explanation for this.

Model	RMSE (gpm)
Persistent forecast	92.6
Default	304.3
$Dt = 0.05\text{h}$	298.1
Z with AB4	299.2
L is evolved	302.1
Interpolate L,Z boundary	327.3
Null L boundary	260.9

Table 1: *Models RMSE comparison.*

5 Conclusions

The reproduction of the first successful NWP highlights both the theoretical strengths and the practical limitations of early computational meteorology. The EBM-QG proved that large-scale atmospheric dynamics could be simulated using a simplified set of physical laws and efficient numerical methods, such as solving the Poisson equation in the frequency domain. However, our numerical results showed significant errors, with a model RMSE of 298.1 gpm compared to a persistence RMSE of 92.6 gpm. This is largely attributed to the strong physical assumptions, the coarse spatial resolution (~ 736 km), and fixed boundary conditions that prevented the proper propagation of features entering the domain.